211.10.02 Buggies. Buggies shall have inflatable rubber tires. Inner surfaces of the bucket shall be smooth with filleted corners, mortar-tight and free of any excessive accumulation of hardened concrete, obstructions or deterioration interfering with the proper discharge of concrete. Grade and ramp-way surfaces shall be sufficiently smooth to prevent segregation of concrete being carried in buggies.

211.10.03 Bottom-dump Buckets and Hoppers. Bottom-dump buckets and hoppers shall have side slopes not less than 60° from the horizontal and shall be equipped with wide free-working and tight-closing discharge gates. The discharge gates shall be constructed to regulate the concrete flow, spring loaded to ensure complete closure and be capable of immediate closure at any time during discharge. Buckets and hoppers shall be cleaned of any accumulation of partially hardened or hardened concrete before and during concrete placement. Gate control mechanisms shall be cleaned and lubricated before being used.

In the use of crane and bucket, segregation of concrete from jarring or shaking shall be prevented. Side-dump buckets shall be used whenever necessary to avoid conflict with extended reinforcing steel or formwork.

211.10.04 Chutes. Chutes shall be of rounded cross section to avoid the accumulation of concrete in corners; be capable of slope adjustment sufficiently steep to permit flow without requiring a slump greater than that specified or required for placement (slope normally required is 1 vertical to 2 or 2½ horizontal).

Baffles and changes in direction may be used to control flow, but not vibration, paddling or water spray.

Chutes or belts shall be rigidly supported but sufficiently mobile to permit discharge as close as possible to the placement location as it progresses. Any long line of chutes or belts shall be covered during hot, dry or windy weather to prevent drying of concrete and excessive slump loss.

211.10.05 Conveyor Operation Rate. Conveyor operation rate shall be dependent on the proper placement and consolidation of the concrete with the belt inclination such that no segregation occurs from any sliding or roll-back of the concrete.

211. 10. 06 Hoppers. Hoppers, set approximately level, shall be supported on specially framed bearers transmitting only vertical loads to the top of the formwork.

211. 10. 07 Pumps - Pumps shall be of sufficient capacity to supply un-segregated concrete, using the design mix proportions at the design slump, to meet the required placement volume at the forms. Satisfactory operation and performance of the proposed pumping equipment with the concrete materials and mixes otherwise suitable and appropriate for the job, along with the line layout and grate size opening, shall be demonstrated to the Quality Manager and Ministry Representative's satisfaction and acceptance.

Any necessary priming of the concrete pump shall be done with a water/cement or water/cement/sand slurry. The prime slurry followed by approximately 0.25 cubic meters of concrete shall not be included in the concrete placement. No other pump aid type admixtures will be permitted.

211.11 Falsework and Formwork.

211.11.01 General. Formwork and falsework shall be designed, supplied, installed and removed in accordance with CAN/CSA-S269.3-M92, CAN/CSA-S269.1-1975 and the requirements for concrete formwork and falsework given in the British Columbia Workers Compensation Act, Occupational Health and Safety Regulations unless otherwise noted.

Formwork and falsework support accessories shall not be welded to the permanent structural steelwork.

Where required, formwork and associated temporary falsework design and drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of British Columbia and experienced in the design and construction of falsework and formwork structures similar to those required to construct this project. No load shall be placed on the formwork or falsework until the professional engineer responsible for the design has inspected and certified, in writing, that the work has been carried out in accordance with the formwork drawings and specifications and a copy of the formwork design engineers' certification has been provided to the Quality Manager and, when requested, to the Ministry Representative.

Bridge deck formwork design shall include a check for the lateral stability of the girders during the placement of deck concrete. Stringers shall be temporarily braced, if necessary, to facilitate deck concreting operations.

Detailed falsework and formwork drawings shall be submitted to the Designer and Ministry Representative for review. The drawings shall be submitted a minimum of two weeks before the start of installation.

211.11.02 Design. All forms shall be designed and built mortar-tight and of sufficient rigidity to prevent distortion due to the pressure of vibrated concrete and other loads incidental to the construction operation. The forms shall be substantial and unyielding, and shall be designed so that finished concrete will conform to the design dimensions and contours. The shape, strength, rigidity, water tightness and surface smoothness of re-used forms shall be maintained at all times. Forms which are unsatisfactory in any respect shall not be used.

All formwork and falsework shall be designed to be completely removable, no stay in place formwork will be permitted.

No dry ties shall be permitted; form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling.

For narrow walls and columns, where the bottom of the form is inaccessible, removable panels shall be provided in the bottom form panel to enable cleaning out of extraneous material immediately before placing the concrete.

211.11.03 Forms for Exposed Surfaces. All forms for exposed surfaces shall be mortar-tight. All exposed edges shall be chamfered by use of a chamfer strip and return corners filleted. Metal bolts or anchorages within the forms shall be so constructed as to permit their removal to a depth of at least 50mm from the concrete surface. Break-back type form ties shall have all spacing washers removed and the tie shall be broken back a distance of at least 50 mm from the concrete surface. All fittings for metal ties shall be of such design that, upon their removal, the cavities which are left will be of the smallest possible size. Torch cutting of steel hangers and ties will not be permitted. Formwork hangers for exterior surfaces of decks and curbs shall be an acceptable break-back type with surface cone, or removable threaded type. Cavities shall be filled with cement mortar and the surface left sound, smooth, even and uniform in color. See DBSSS 211.17 for details on finishes, mortar mixes and other relevant information.

211.11.04 Deck Formwork. Formwork for decks, curbs, sidewalks and parapets shall be fabricated so that the lines and grades shown on the Design are achieved. For suspended slab bridge decks the formwork shall be designed to allow for an adjustable haunch to facilitate the casting of a constant depth deck slab as shown on the Design. Girders will be erected to normally accepted standards of tolerance; it shall be necessary to adjust the formwork to compensate for camber, variances in girder dimensions, positioning, and sweep.

Prior to commencing deck formwork and subsequent to the completion of the erection of girders, with the girders in a thermally neutral condition and with no dead load other than the dead load of the girders and necessary fall protection, the Contractor shall profile all the girders. Camber elevations are to be taken at 1000 mm centres maximum (or as indicated on the Design), centrelines of all bearings and ends of girders. The camber elevation locations are to be laid out to an accuracy of 25 mm horizontally and camber elevations taken to the nearest 1 mm vertically. The Contractor shall supply the camber elevations to the Designer. The Designer shall calculate and provide the Contractor with the haunch heights for setting the deck slab soffit, and the design finished grade elevations for the deck, complete with anticipated dead load deflections at 1000 mm stations along the centreline of roadway.

In the event that actual girder camber values vary significantly from the design values, the Contractor will be required to modify the original design finished grade as directed by the Designer.

211.11.05 Removal of Falsework and Formwork. Where formwork and associated falsework have been engineered by a professional engineer, forms and their supports shall not be removed without the approval of the professional engineer responsible for the formwork and falsework design and without the acceptance of the Designer. In determining the time for the removal of falsework and formwork, consideration shall be given to the location and character of the structure, the weather and other conditions influencing the curing of the concrete, and the materials used in the mix.

Supports shall be removed in such a manner as to permit the concrete to uniformly and gradually take the stresses due to its own weight together with any construction loads likely to be imposed. All formwork must be removed from the completed structure.

211.12 Construction Joints.

211.12.01 General - Definition. Joints between adjacent or successive lifts of concrete where the previously cast concrete cannot be plasticized with a vibrator, or any joint required by the Contractor's method of construction. Construction joints generally shall be in accordance with the following requirements:

- kept to a minimum
- concealed by a natural break or line in the structure
- provided for by the formwork design and placing techniques.

Emergency construction joints shall conform as nearly as possible to planned joints. In any case, concrete must be levelled as well as possible and, if lifts are partial, a vertical bulkhead shall be installed. Permission from the Designer and Ministry Representative shall be required to complete the cast following any emergency joint installation.

211.12.02 Joint Edges. Joint edges shall be preformed whenever possible for a true line upon form removal. Pour-strip shall be positioned relative to form ties so as to minimize form bulge at the bottom of the subsequent cast, but concrete may be cast to the top of a form instead of the pour-strip. Concrete surfaces outside of the reinforcing steel shall be steel trowelled to a true line.

211.12.03 Surfaces Preparation. Joint surfaces of the hardened concrete shall be thoroughly cleaned of foreign

matter and laitance to present clean, sound surfaces with the coarse aggregate partially exposed and achieved by:

- brushing immediately following initial set
- treatment with a surface set retardant, followed by green cutting
- sand blasting
- water blasting.

Where roughening of the hardened concrete is specified, the surface shall be roughened to a full amplitude of at least 5mm.

211.12.04 Bonding of Joints. Bonding of joints shall be achieved through the use of a water-cement slurry or, if specified, an epoxy-type bonding agent conforming with ASTM C 881M applied in strict accordance with the manufacturer's specifications. Slurry, when used, shall have a water-cement ratio equal to that of the concrete being placed and be brush applied to a 3 to 5 mm thickness for 100% coverage on to the cleaned existing concrete surface within five minutes of placing concrete.

211.13 Consolidation.

The equipment specifications in DBSS 211.13.01 to DBSS 211.13. 02 inclusive are guidelines only, and as such, are not mandatory. DBSS 211.10.03 is mandatory. Consolidation shall also meet the requirements of CSA A23.1.

211.13.01 Vibrators – General. The number, type, design and operation of vibrators will be subject to the acceptance of the Designer and Ministry Representative.

Vibrators to have the following capabilities and requirements:

- diameter and frequency of vibrators to conform to Table 211-L
- vibration transmission to the concrete at frequencies not less than 130 Hz
- vibration intensity sufficient to be visually observed on concrete with not more than 25 mm slump over a radius of at least 450 mm
- sufficient number of vibrators available to properly compact each 8 m³ batch within 15 minutes after it is placed in forms, with at least two vibrators provided for each concrete placing unit
- stand-by vibrator and generator available on the job at all times in case of breakdown.

211.13.02 Mechanical Vibrators. Mechanical vibrators to thoroughly consolidate concrete immediately after placing.

Vibrator usage to be as follows:

- not inserted between reinforcing steel and formwork nor directly on to the reinforcing steel
- inserted vertically at a rate of 75 mm per second and penetrating into the preceding layer at least 50 mm
- withdrawn at a rate of 150 mm per second
- grid pattern movement starting 100 mm from a formed surface and at a maximum spacing of 300 mm thereafter.

211.13.03 Form Vibrators. Form vibrators may be permitted at the discretion of the Designer and Ministry Representative where rebar configuration would prohibit the use of internal vibrators.

TABLE 211-L REQUIREMENTS FOR DIAMETER AND FREQUENCY OF VIBRATORS

DIAMETER (mm)	FREQUENCY (Hz)	RATE OF PLACEMENT OF CONCRETE PER VIBRATOR (m^3/h)	APPLICATION
20-40	170-250	1-4	very thin members
30-60	150-225	2-8	walls and slabs 200 mm - 300 mm
50-90	130-200	5-15	larger members

211.14 Finishing.

211.14.01 Finishing Top Surfaces - Plastic concrete, struck off immediately following casting and before the accumulation of bleed water, shall be finished by hand or mechanical floating with the least possible disturbance. Water, cement or proprietary topping material shall not be added to the surface during finishing.

No steel floats or trowels shall be used on exposed surfaces.

211.14.02 Strike-off and Finishing Machines - Strike-off and finishing machines for surfacing concrete such as slabs on grade, pavements, etc. shall conform to the following:

- capable of accurately finishing concrete surfaces to the grade and elevation shown on the Design
- surface finish production free of open texturing, plucked aggregates or projections
- maintained in good mechanical repair and free of hydraulic fluid leaks, oil leaks, excessive grease and other contaminants.

211.14.03 Trowel Finishing - Trowel finishing where required:

- by the Design,
- for unformed surfaces exposed to view from a close distance,
- by close tolerance requirements (such as bearing surfaces),

shall be commenced after bleed waters have disappeared to reduce the texture of the concrete surface and produce a smooth blemish-free surface.

211.14.04 Other Textured Finishes - Other textured finishes shall be applied as and where specified.

211.14.05 Damage or Scaling - Any damage to or scaling of surfaces shall be repaired to the satisfaction of the Quality Manager and Ministry Representative.

211.15 Curing.

211.15.01 General. Freshly deposited concrete shall be protected from freezing, abnormally high temperatures or temperature differentials, premature drying, excessive moisture, moisture loss, heavy shocks, excessive vibrations and high stresses, for the period of time necessary to develop the desired properties of the concrete.

The curing period will begin following the placement of the concrete in the section cast. It shall be the responsibility of the Contractor to ensure that the system of curing and protection is properly planned, constructed and maintained throughout the entire curing period.

211.15.02 Acceptable Methods and Procedures. Acceptable curing methods and procedures shall include the following:

- concrete surfaces shall be cured for a period of seven consecutive uninterrupted twenty-four hour periods, however, if the Designer has accepted the use of high-early strength cement(HE), this period may be reduced as recommended by the Designer and as permitted by the Ministry Representative.
- surface temperature maintained between 10°C and 35°C, unless high-temperature curing (i.e., by hot blankets or steam) is acceptable to the Designer and Ministry Representative.
- hot weather and cold weather special requirements are covered by DBSS 211.18 and DBSS 211.19. The determination and recording of air and concrete temperatures to verify compliance with the requirements for hot and cold weather concreting shall be the responsibility of the Contractor. The location and methodology for the determination and recording of temperatures shall be acceptable to the Quality Manager and Ministry Representative.
- formed concrete protection, with form-work left in place during the curing period (usually seven days), is generally not necessary except during extremes of hot or cold weather temperatures. Where formwork is removed before the end of the curing period, the concrete shall be protected in the manner specified below for unformed surfaces during the remainder of the curing period
- unformed surface protection using burlap and water or where permitted, other moisture-retaining methods considered adequate to promote curing, shall include the following common methods and requirements:

i) Burlap - Soaking wet burlap of acceptable quality shall be carefully laid on the surface of the concrete as soon as the concrete has set sufficiently to support the burlap without marring the surface. Strips shall be overlapped 150 mm, adequately held down and maintained in place in a thoroughly wet condition throughout the curing period (usually seven days). Wetness of burlap shall be maintained at all times.

ii) Vapour Barrier - Vapour barrier of acceptable quality of opaque white-pigmental 0.10 mm thick ("4 mil") polyethylene sheet, white upper surfaced paper or other permitted material shall be of such size and placement to minimize the number of laps. The material shall be laid, lapped and held down around the edges and at the laps:

- to prevent displacement

- to provide and maintain an effective vapour barrier
- prevent any flow of air between the vapour barrier material and the concrete
- kept in place throughout the curing period (usually seven days).

However, the barrier material shall not be pulled tight against the fresh concrete surface so as to smooth the required surface texture.

211.15.03 Removal of Protective Materials. Protective materials shall be removed entirely from all concrete surfaces at the end of the curing period and the surfaces left clean.

211.16 Dimensional Tolerances. Dimensional tolerances shall be as shown in Table 211-M.

NOTE: The tolerances listed in Table 211-M are acceptable only in so far as they do not prevent the proper fit of structural members.

Table 211-M DIMENSIONAL TOLERANCES

Trueness of bearing surfaces	such that there is no gap over 1 mm under a 1 m straight edge
Trueness of bridge deck surfaces	such that there is no gap over 3 mm under a 3 m straight edge
Trueness of slope pavement surfaces	such that there is no gap over 12 mm under a 3 m straight edge
Trueness of other surfaces exposed to view	such that there is no gap over 12 mm under a 3 m straight edge
Curbs and parapets	such that there is no gap over 3 mm under a 3 m straight edge
Horizontal distance to concrete surfaces	±50 mm
Elevations of concrete surfaces except bearings and decks	±25 mm
Elevations of bridge deck surfaces	±15 mm
Elevations of bearing surfaces	±5 mm
Variation from plumb	1:400, but not more than 30 mm
Dimensions of members over 10 m	±25 mm
Dimensions of members under 10 m	-6 mm, +15 mm
Location of anchor bolts	± 6 mm
Projections of anchor bolts, other embedments	±13 mm

211.17 Finishing Formed Surfaces.

211.17.01 Class 1 Finish. Class 1 Finish is the basic finish to be produced on all formed surfaces not exposed to view unless a better finish is specified or required by the Design.

In order to produce a Class 1 finish, the formwork shall be mortar tight. Panel marks and texture are of no importance.

All ties, bolts, nails and other metal specifically required for construction purposes shall be removed or cut back to a depth of 50 mm from the surfaces of the concrete and the resulting holes filled.

No dry ties shall be permitted; form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling.

Honeycombs and voids over 500 mm² in area shall be filled. Honeycombs and voids shall not be repaired until inspected by the Quality Manager and Ministry Representative as special methods of repair may be required where occurring in structural elements. Otherwise, concrete surfaces shall be repaired as follows:

- remove all porous concrete by cutting 20 mm into sound concrete
- thoroughly clean reinforcing steel without damaging same
- cut edges of cavities as nearly perpendicular to the concrete surface as possible and deep enough to hold the patching mortar
- keep surrounding concrete saturated for 30minutes before patching
- after the sheet of free water has disappeared from the concrete surfaces, apply a latex bonding agent, in strict accordance with the manufacturer's instructions,
- patch with mortar, proportioned as specified in DBSS 211.17.05 and mixed with the minimum amount of water necessary to make a workable paste: for voids deeper than 10 mm, multiple layers of mortar each with a maximum thickness of 10 mm shall be used with 30 minute intervals allowed between layers
- trowel the surface of the patch flush with the surface of the surrounding concrete.

211.17.02 Class 2 Finish. Class 2 Finish is to be produced on all formed surfaces exposed to view from a moderate distance, such as surfaces of abutments and piers, and to any surfaces for which a Class 2 finish is specified or required by the Design. A Class 2 finish shall provide surfaces of uniform colour and texture as viewed from 25 m.

In order to produce a Class 2 finish, formwork shall be mortar tight and shall render a true surface. Fins 3 mm wide (maximum) shall be allowed at the panel joints; however, sheathing joints must be mortar tight. Irregularities of 3 mm in height with areas of 50 mm x 75 mm shall be allowed to a maximum of four such areas per 3 m² of formwork. Patches of dissimilar material will not be permitted. Horizontal and vertical joints shall be aligned.

All ties, bolts, nails and other metal specifically required for construction purposes shall be removed or cut back to a depth of 50 mm from the surfaces of the concrete and the resulting holes filled.

No dry ties shall be permitted; form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling.

Honeycombs and voids over 25 mm diameter shall be filled, and all bugholes over 5 mm diameter shall be pointed. All fins and projections shall be removed with a hand stone or power grinder. The use of a power grinder shall be kept to a minimum and confined to the areas required.

When a rubbed finish is not called for, patches shall be textured with a mortar float or lightly brushed after trowelling smooth.

Where more than 50 voids or bugholes over 5 mm diameter occur per square metre, or if the surfaces are not acceptably uniform in colour or texture, the entire area affected shall be given a rubbed finish, as follows:

- allow patches to attain an initial set
- keep surfaces saturated with water for 60 minutes before applying mortar
- remove free water from surfaces
- apply mortar, proportioned as specified in DBSS 211.17.05 and mixed at least 60 minutes before application, to the concrete surfaces with a sponge, float or a hand carborundum stone, and work mortar well into the surfaces
- allow mortar to dry 30 - 60 minutes
- remove residues from the surfaces by rubbing with clean dry burlap but with no addition of neat cement to the surface during the "sacking" operation and finally cure the surfaces as specified in DBSS 211.15.

211.17.03 Class 3 Finish. Class 3 Finish is to be produced on all formed surfaces exposed to view from close distances, such as surfaces of curbs and parapets, and to any surfaces for which a Class 3 finish is specified or required by the Design. A Class 3 finish shall provide surfaces of uniform colour and texture when viewed from less than 15 m.

In order to produce a Class 3 finish, the formwork shall render a true smooth surface, free from fins and projections. New plywood or steel is necessary to produce the required finish. Re-use of plywood forms will be permitted only if in an "as new" condition. Repairs to the forms shall be with full panels of sheathing only.

All ties, bolts, nails and other metal specifically required for construction purposes shall be removed or cut back to a depth of 50 mm from the surfaces of the concrete and the resulting holes filled.

No dry ties shall be permitted; form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling.

Patching shall be:

- patch installation projecting slightly from the surrounding concrete surface
- after 24 hours, saturated and carefully finished to render the surfaces of the patch true to the surrounding concrete, but any grinding using a hand carborundum stone kept to a minimum

When a rubbed finish is required, the applied mortar shall be carefully worked into the surface area being finished and, using a hand carborundum stone manipulated in a circular motion, the entire surface rendered true and smooth with all excess mortar removed.

211.17.04 Class 3 Finish Alternative Procedures. Class 3 Finish alternative procedures may be used when, in the opinion of the Ministry Representative, the formwork may safely be removed at an early age and the concrete finished while still partly plastic. Procedures shall include the following:

- careful removal of the formwork when the concrete has sufficiently hardened so as to hold its shape
- finishing the surfaces immediately (within 4 to 8 hours of casting, depending upon weather)
- careful rubbing of the concrete surfaces with a carborundum stone, occasionally wetting same to produce a paste from the surface of the concrete
- rubbing continued, adding necessary sand-cement mortar in small quantities to fill voids, until all voids and form marks are removed
- surface curing as specified.

Modifications to the surface texture shall be achieved by grinding or other means.

211.17.05 Patching Mortar. Patching mortar where specified above shall be constituted as follows:

Cement: 60% Normal Portland
40% White Normal Portland

Liquid: 70% Water
30% Latex Bonding Agent

Sand: Passing a 1.25 mm sieve

NOTE: Cement blend above may be adjusted to provide close match to the concrete colour.

211. 18 Hot Weather Concreting.

When ambient air temperature is 25°C or higher or there is a probability of it rising to 25°C during the placing period (as forecast by the nearest official meteorological office), hot weather concreting procedures as outlined below, shall be used, for all concrete other than deck concrete. Additional requirements for hot weather concreting for bridge decks including diaphragms, bridge parapets, bridge medians, bridge sidewalks and approach slabs are given in DBSS 413, Bridge Decks and Concrete Overlays.

Prior to the placement of concrete, the Contractor shall submit to the Quality Manager and Ministry Representative for acceptance, the proposed placing operations complying with the requirements for hot weather concreting.

Curing shall be accomplished by water spray or by using saturated absorptive material, such as burlap.

Formwork, reinforcement and concreting equipment shall be protected from the direct rays of the sun or cooled by fogging and evaporation.

Concrete for piers, abutments and footings of bridges shall not have a temperature greater than 25°C at any time prior to placement and curing.

The initial temperature of concrete prior to placement may be estimated from the temperatures of its ingredients by using the following equation:

$$T = \frac{0.22(T_a M_a + T_c M_c) + T_w M_w + T_{wa} M_{wa}}{0.22(M_a + M_c) + M_w + M_{wa}}$$

where T_a, T_c, T_w and T_{wa}

= temperature in °C of aggregates, cementing materials, added mixing water, and free water on aggregates, respectively

where M_a, M_c, M_w and M_{wa}

= mass in kilograms of aggregates, cementing materials, added mixing water, and free water on aggregates, respectively

If deemed necessary, the Contractor shall employ the following hot weather concreting procedures to reduce the concrete temperature:

a) Mixing Water

- Shading of storage tanks and water supply lines;
- Refrigeration of mixing water; and/or
- Adding cubed, shaved or chipped ice directly to the concrete mixer drum, and ensuring ice is of a particle size to be completely melted before the concrete is discharged from the mixer drum for placement.

b) Aggregate

- Shading of coarse and fine aggregate stockpiles; and/or
- Wetting of coarse aggregate and allowing time for drainage of free water.
- Note that wetting down of stockpiles of fine aggregate is not permitted.

c) Mixing: Mixing time should be kept to a minimum, allowing adequate time for complete mixing of the concrete. The external surface of the mixing drum should be frequently wetted down.

d) Placing: Production of concrete during hot weather shall be scheduled for a continuous supply of concrete at the Site.
Adequate manpower, equipment and standby provisions shall be provided in order to place and consolidate the delivered concrete within 45 minutes after initial mixing. Further, the time interval between placing batches shall not exceed 30 minutes.
Time of placement extensions during hot weather concreting may be considered by the Ministry Representative and may require the use of Hydration Stabilizing Admixtures (HSA's) conforming to the requirements of ASTM C494 Type B, Retarding or Type D, Water-Reducing and Retarding Admixtures. Use of HSA's shall be in accordance with DBSS 211.08.02.

For large unformed surfaces, the quantity and rate of placing of concrete shall be dependent upon the rate of finishing and climatic condition.

When the rate of evaporation from unformed surfaces as estimated from Drawing SP211-01 "Surface Evaporation Rate" exceeds 1.0 kg/m²/hr, either concrete operations shall cease or, upon acceptance of the Quality Manager and Ministry Representative, necessary precautions taken to prevent plastic shrinkage as detailed in CSA A23.1 Section 7.4.2.2.

e) Curing during hot weather shall be commenced as soon as possible.

Forms shall be kept moist and loosened as soon as this can be done without structural damage, with a flow of water introduced and allowed to run down inside.

Newly exposed concrete surfaces shall be protected from premature drying by the application of pre-wetted curing materials or a continuous water spray as the forms are removed.

Large exposures of concrete require special precautions as follows:

- burlap application commenced as soon as the surface will support the mass without excessive deformation, with this surface kept continuously wet by fog spraying until a waterproof barrier is securely in place
- concrete under the curing blanket is not allowed to dry out for a minimum of seven days from the time of initial covering, necessitating frequent checking and soaking by the addition of water under the top evaporation barrier
- wet curing followed by the removal of the evaporation barrier only under such conditions that the wetted burlap is permitted to dry slowly before removal and the surface is exposed to prevent excessive thermal shock.

211.19 Cold Weather Concreting. Cold weather concreting procedures shall be used when the ambient temperature is, or is forecast to be, below 4°C during placement, and/or is forecast to fall below 4°C during the first seven days after placing. The Contractor shall be fully responsible for the protection of concrete during cold and adverse weather conditions and shall maintain a minimum concrete temperature of 10°C for a minimum of seven continuous days. Prior to the placement of concrete, the Contractor shall submit to the Quality Manager and Ministry Representative for acceptance, the proposed placing operations complying with the requirements for cold weather concreting. The following provisions for cold weather concreting shall apply:

a) All aggregate and mixing water shall be heated to a temperature of at least 20°C but not more 65°C. However, the temperature of the combined water and aggregates shall not exceed 40°C when combined with the cement. The temperature of the concrete shall be between 10°C and 25°C at the time of placement.

b) Aggregates shall be heated to eliminate frozen lumps, ice and snow without overheating or excessive drying. Aggregates shall not be heated above 65°C, and all lumps of frozen aggregate shall be excluded from the mix.

c) Formwork, Reinforcing Steel: Before any concrete is placed, all ice, snow or frost shall be completely removed from the forms and the temperature of contact surfaces

raised to a minimum of 5°C, with such minimum established and maintained for at least one hour prior to placement.

Heat shall be applied uniformly and at a rate which will not induce excessive thermal stresses in the section being heated. Ambient air temperatures shall not exceed concrete temperatures by more than 13°C and shall be raised at a maximum rate of 2°C/hr.

Projecting reinforcing steel shall be insulated when temperatures are between 0°C and -10°C for a minimum of 300 mm away from the fresh concrete and for a minimum of 600 mm when temperatures are below -10°C.

d) Other Contact Surfaces: Concrete shall not be placed on frozen surfaces except in pile founded footings where a minimum layer of 25 mm rigid insulation shall be installed. Rock surfaces shall be heated by forced air. Heating shall continue until all standing frost crystals are dissolved. Frozen embankment shall be thawed and recompacted prior to placing concrete thereon. The temperature of existing concrete work shall be raised to a minimum of 5°C and maintained at this temperature for a time sufficient to raise the entire mass of concrete to this temperature.

e) Curing and Protection necessary to maintain a minimum concrete temperature (10°C) shall be provided throughout the curing period.

Heated enclosures shall be kept at 95% minimum relative humidity. Concrete surfaces shall be protected by formwork or impermeable membranes from direct exposure to the combustion gases of heating. The use of salamanders, coke stoves, oil or gas burners and similar spot heaters which have an open flame and intense local heat is prohibited without the Quality Manager and Ministry Representative's acceptance. The enclosed air temperature shall not be less than 10°C nor more than 26°C.

Where practical, insulated forms, capable of maintaining the surface of the concrete at not less than 10 °C for a period of seven days, may be used instead of enclosures and heating. If forms are insulated, exposed horizontal surfaces shall be protected with a similar layer of the insulating material securely fastened in place. If the insulated forms do not maintain the proper temperature at the concrete surface, auxiliary protection shall be used to provide additional heat as described above.

The determination and recording of air and concrete temperatures to verify compliance with the requirements for cold weather concreting shall be the responsibility of the Contractor.

The Contractor is responsible for furnishing and using a 24-hour minimum/maximum or continuous temperature recording thermometer to record the air temperature within the enclosure and/or insulated forms. Daily records shall be submitted to the Quality Manager and to the Ministry Representative upon request.

At the end of the curing period, heating and protection shall be withdrawn in such a manner so as not to induce thermal shock stresses in the concrete. The temperature of the concrete shall be gradually reduced at a rate not exceeding 10°C per day to that of the surrounding air. To achieve this, in a heated housing, the heat shall be slowly reduced and then shut off, and the whole housing allowed to cool to air temperature before the housing itself is removed. However, the protection shall not be removed until the temperature of the concrete has fallen to within 10°C of the temperature of the outside air.

211.20 Underwater Concreting. Underwater concreting procedures shall be used when structures, or parts of structures, are to be constructed under water. The concrete, methods of placement either by pumping or tremie pipes, and equipment shall be in accordance with the following, or as otherwise directed by the Designer and Ministry Representative:

a) Proposals outlining procedures shall be submitted for review to the Designer and Ministry Representative 3 weeks prior to placement of concrete. Placement shall comply with the provisions of the proposal.

b) Materials shall conform to the requirements of DBSS 211.04, except that coarse aggregates shall contain not more than 25% angular particles by mass.

c) Mix proportions shall be as specified in DBSS 211.03, except:

- cement content not less than 400 kg/m³
- water-cement ratio no greater than 0.40
- slump of 150 mm ±20 mm
- admixtures used to achieve plastic, cohesive and flowable concrete if acceptable to the Quality Manager and Ministry Representative
- retarding type admixture if accepted by the Quality Manager and Ministry Representative shall be used to prevent the formation of cold joints with the lift method of placement or where the concrete surface area is large.

d) Heating of water or aggregates or both, to obtain a suitable placing temperature, shall be done in accordance with DBSS 211.19. Concrete, when mixed, shall have a temperature of between 15°C and 25°C and shall not be placed in water having a temperature less than 2°C.

e) Cofferdams or forms into which concrete is to be placed shall be sufficiently tight to prevent the loss of mortar and, if in running water, reduce the velocity of flow within the cofferdam or forms to not more than 3 m per minute. Dewatering will not be permitted until at least 24

hours after concreting has been completed.

f) Pumping and tremie pipe requirements shall be as follows:

- length sufficient to reach the lowest point of deposit but held 150 mm to 250 mm above the bottom of the placement
- submerged into the placed concrete and held in a near vertical position at all times
- supported in such a manner to allow vertical movement
- discharge at locations not farther apart than 7 m
- discharge end fitted with a 3 m section of steel pipe
- fed by either chutes or pumps with a continuous flow of concrete.

g) Priming of tremie pipes shall be by inserting a plug of burlap, 25 mm less in diameter than the tremie pipe, in the top of the pipe and carefully controlling concrete flow as the plug passes through the pipe. Where priming is lost, the priming procedure shall be repeated.

h) Concrete flow, once begun, shall be continuous through the pipe, at a minimum of 15 m^3 of concrete supplied to each tremie or pump per hour, but controlled so as not to overflow the feed hopper. The flow of concrete shall be continuous to the end of the batch and the pipe lowered in the placed concrete to maintain its prime.

i) Placement of concrete underwater shall be either to the full depth of placement or in lifts and shall conform to the following requirements:

- seals shall be placed by beginning at one end and continuing until final grade is achieved
- point of placement shall be relocated by approximately 3 m, either laterally or longitudinally and placement continued to grade
- above procedures repeated to the completion of the seal or when the lift method of placement is used
- flow of concrete shall be regulated to produce approximately horizontal surfaces
- each lift placed before the preceding lift has taken initial set
- depth and area of each lift dependent upon the rate of placement.

Concrete in its plastic state shall not be disturbed either directly or indirectly by puddling or vibration.

Should interrupted placement be sufficient for the concrete to take initial set, the laitance shall be removed from the concrete surface before concreting is resumed, but with the resumption of concreting delayed indefinitely, laitance shall be removed not later than 36 hours after the interruption.

SECTION 213

TIMBER BRIDGES - CONSTRUCTION

Note: Timber bridges may be used only when explicitly identified in the Contract as an acceptable product.

213.01 Materials - All materials shall be of the quality, grade and finish as shown on the Design and shall conform to the requirements of the DBSS 903, DBSS 904, DBSS 905 and DBSS 908 that are relevant.

213.02 Logs for Cribs and Log Stringers - Logs for cribs shall be cut from live trees and shall be peeled before use.

Stringers shall be cut from live trees not less than 30 days before use and shall be seasoned with the bark on. Immediately before use in the work, all bark shall be peeled and the log trimmed smooth of all knots and projections.

213.03 Work Practice - The standard of work shall be first-class throughout and in accordance with the best modern practice.

213.04 Handling and Storing Materials - All materials shall be handled with reasonable care. Timber shall not be bashed, cut or punctured or otherwise damaged.

Dogs, hooks, peavies or other equipment shall not be used on the side surfaces of treated timber. All handling of treated timber with pointed tools shall be confined to end grain and shall be such as to avoid damage of the original pressure-treated surface.

Subject to the approval of the Quality Manager and Ministry Representative, the Contractor shall make good any superficial damage of treated timber by the methods outlined in the current edition of C.S.A. 080 Specification for Wood Preservation. Timber which, in the opinion of the Quality Manager and Ministry Representative, cannot be made good by such methods will be rejected.

Minor damage to untreated timber shall be repaired and severely damaged pieces will be rejected.

When unloading at the site of the work, the various pieces shall be sorted and stacked in such a manner as to require a minimum of rehandling before being placed in the structure.

Level ground cleared of snow, brush and weeds shall be provided for stacking the timber. The timber shall be placed on blocking to raise it above the ground.

213.05 Framing - Framing shall be first class in every respect and performed by skilled workers only. No open joints, blocking or shimming of joints will be allowed. Daps, gains and dove-tails in trestles, culverts, cribs, etc. may be finished with an adze, but all joints shall go together without pounding or excessive cramping up and have full bearing in contact.

Holes shall be bored with an auger to the sizes shown in Table 213-A.

Spikes that are likely to crack or unnecessarily damage the timber shall have holes bored 1/16" small.

213.06 Field Treatment - Unless so specified, treated timber shall not be cut or dapped except for cutting to length. Cut ends and specified cuts and daps of treated timber (except piles) shall be given three coats of the same preservative used in the treatment; creosote, if used, shall be hot.

The heads of all treated piles, after cut-off, shall be covered with alternate layers of hot roofing pitch and loosely woven fabric using four applications of pitch and three layers of fabric.

The fabric shall measure at least six inches more in dimension than the diameter of the pile and shall be neatly folded down and secured by large-headed galvanized roofing nails or by binding with not less than seven complete turns of galvanized wire securely held in place by large-headed galvanized roofing nails or staples. The edges of the fabric projecting below the fastening shall be neatly trimmed.

The heads of all untreated piles, after cut-off, shall be thoroughly swabbed with three coats of hot creosote oil and if they are to be capped, they shall also be coated with hot roofing pitch.

TABLE 213-A HOLE SIZES

	FIR AND HARD-WOODS	CEDAR AND SOFT-WOODS
BOLTS	1/16" Large	Same size as bolt
DRIFT BOLTS	Same size as bolt	1/16" Small
DOWELS	1/16" Large for ends of posts, but in sills make same size as dowel	1/16" Large for ends of posts, but in sills make same size as dowel

All bolt holes drilled in the field in treated timber shall be treated with hot creosote oil applied with an approved pressure bolt hole treater.

All surfaces in contact and all bolt holes in untreated timber shall be swabbed with a water-borne preservative.

213.07 Preparation of Laminated Decks for Wearing Surface

213.07.01 For Asphalt Plank - Depressions in laminated decks shall be levelled by placing and compacting a fine sand levelling course.

The levelling course shall consist of a mixture of fine durable sand conforming to the gradation shown in Table 213-B.

The sand shall be thoroughly mixed with asphalt emulsion, Type SS-1 which shall conform to the requirements of DBSS 951.17. The asphalt emulsion content shall be such that an air void content of 5% to 8% in the compacted mixture is obtained. This will normally require an asphalt emulsion of 15% to 25% by weight of the total mixture.

Before laying and nailing asphalt plank, laminated decks shall be swabbed with oxidized mopping asphalt Type 2 conforming to the requirements of Table 213-C.

213.07.02 For Asphalt - Prior to placing asphalt wearing surface, creosote-treated laminated decks shall be lime-scrubbed where creosote oil has "bled" to the surface.

To prevent "bleeding" of the creosote oil, the mix temperature of hot-mix asphalt placed on these decks shall be as low as is practicable.

213.08 Not Used

TABLE 213-B CONSISTENCY OF THE LEVELLING COURSE

U.S. STANDARD AND ASTM SIEVE SIZES	PERCENTAGE PASSING BY WEIGHT (%)
#8	100
#16	100 - 60
#30	80 - 46
#50	62 - 35
#100	46 - 25
#200	35 - 18

TABLE 213-C REQUIREMENTS FOR OXIDIZED MOPPING ASPHALT TYPE 2

TEST DESCRIPTION	ASTM TEST METHOD	TYPE 2	
		MINIMUM	MAXIMUM
Softening Point (R & B), ^{o}F	D36	165	175
Flash Point, C.O.C., ^{o}F	D96	450	---
Penetration @ 32°F	D5	12	20
Penetration @ 77°F	D5	20	35
Penetration @ 115°F	D5	---	70
A.P.I. Gravity @ 60°F	D71	4.6	5.8
Ductility @ 77°F, 5 cm minimum	D113	3	---
Loss on Heating to 325°F, 5 hr., %	D6	---	0.3
Penetration of Residue, % of original	D5	80	0
Total Bitumen (soluble in CS_2) %: Mineral Stabilized Asphalt	D4	80	---
Total Bitumen (soluble in CS_2) %: Asphalt Without Mineral Stabilizer	D4	99	---
Proportion of Bitumen Soluble in CCl_4		99.5	---

SECTION 214

TIMBER PILING CONSTRUCTION

Note: Timber piles may be used only when explicitly identified in the Contract as an acceptable material.

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

214.01 Materials - All piling shall conform to requirements of DBSS 906, Round Timber Piles, and where specified to DBSS 908, Preservative Treatment.

214.02 Pile Lengths - Piles shall have minimum lengths in accordance with those listed on the Design. Allowances will be made for such additional lengths of pile as may be required to provide fresh heading and to suit the method of operation.

Pile lengths determined by the Design, are those expected to remain in the completed structure plus 600 mm for cut-off, except for end bearing piles where the allowance for cut-off and bearing depth irregularity is as determined by the Designer.

214.03 Handling of Piles - The use of dogs, hooks, peavies or any other equipment which may damage the round surface of a treated pile will not be permitted.

Pointed tools used in the handling of treated piles shall be confined to end grain.

During lifting, long piles shall be supported at a sufficient number of points, properly located, to prevent damage due to excessive bending.

Subject to the approval of the Quality Manager and Ministry Representative, the Contractor shall make good any superficial damage of treated piles by the methods outlined in the current edition of CSA 080 Specifications for Wood Preservation. Piles damaged by the Contractor, which, in the opinion of the Quality Manager and the Ministry Representative, cannot be made good by such methods, will be rejected and replaced.

214.04 Hammer - Piles may be driven with a drop hammer or a mechanical hammer. Drop hammers shall weigh not less than 1,400 kg for piles less than 15 m long and not less than 1,600 kg for piles 15 m and over in length, and in no case shall the weight of the hammer be less than the combined weight of driving head and pile. Mechanical hammers shall develop a total energy not less than 3,900 kg per blow.

214.05 Leads - Pile drivers with fixed leads or with hanging or swinging leads that can be held in a fixed position during driving shall be used to drive all piles.

Batter piles shall be driven using leads adaptable to their driving.

When driving treated timber piles, the use of spuds and chocks in the leads shall be kept to a minimum in order that the protective treatment will not become bruised or broken.

214.06 Preparation for Driving

214.06.01 Driving Caps and Bands - The heads of timber piles, when driving is such as to unduly damage them, shall be protected by driving caps acceptable to the Designer, preferably having a rope or other suitable cushion next to the pile head and fitting into a casting which in turn supports a timber shock block. Alternatively, the pile head may be banded with a pile ring and covered with a steel wire mat. When driving treated piles, a driving cap shall be used.

214.06.02 Pointing - Where soil conditions, in the opinion of the Designer, require the pointing of piles, the Contractor shall fit the piles with steel points.

Steel points fabricated in accordance with SS Drawing SP214-01, or as detailed by the Designer, shall be supplied by the Contractor.

214.06.03 Excavation - Footing excavation and seal excavation, where applicable, shall be completed with any necessary allowance made for upheaval before the driving of footing piles. Should too great an allowance be made, backfilling with gravel will, in general, be permitted to raise the bottom of the excavation to the correct elevation. Any material forced up between the piles to above the footing level shall be removed to the correct elevation.

214.07 Driving - Unless otherwise specified by the Design, all piles shall be driven with the small end downward. The fall of a drop hammer shall be regulated to avoid damage to the piles and under no circumstances shall exceed 3 m.

Each pile shall be square-headed, properly pointed, and set fair in the leads; if it becomes split or broomed on the top it shall be fresh headed.

For drop hammers:

$$s = \frac{wh}{6p} - 0.03$$

For mechanical hammers:

$$s = \frac{wh}{6p} - 0.003$$

Where: s = average penetration per blow in metres under last 10 blows
w = weight of moving part of hammer in newtons
h = effective fall of hammer in metres
p = allowable load capacity of pile in newtons

Piles shall be driven to the minimum depth indicated on the Design and unless otherwise ordered by the Designer, to practical refusal, which shall be to the depth at which the piles have a load carrying capacity of 20 tonnes and an average penetration per blow determined by the above formulae:

Where it is required by the Designer to penetrate a thin stratum (which may show practical refusal in driving) to obtain greater depth in a less resistant formation, driving shall be continued through the stratum regardless of the definition laid down for practical refusal.

Any pile which, in the opinion of the Designer or Ministry Representative, is so damaged in driving as to be unfit for the use for which it is intended, or is so far out of position, or off vertical or designated batter, as to require excessive force or manipulation to restore it to its proper position, shall be withdrawn and replaced by a new pile. If this is not possible, a new pile shall be driven adjacent to the defective pile.

Where boulders or other large obstructions make it impossible to drive piles in the location shown on the Design, the Designer may direct the Contractor to drive an additional pile, or piles.

214.08 Additional Equipment - Where, in the opinion of the Designer or Ministry Representative, the bounce of the hammer is excessive or where the required depth or resistance is not obtained by the use of a hammer complying with the above minimum requirements the Contractor shall provide a heavier hammer to complete the work.

214.09 Accuracy of Driving - All piles shall be driven in such manner that any variation from the vertical, or batter as shown on the Design, shall not exceed 20 mm/m. Piles in trestle bents shall be so driven that the cap may be placed in its proper location without including excessive stresses in the piles. Cutting, springing, dapping or facing-off for placing bracings and planking on piles will not be permitted.

After driving, footing piles shall not be more than 75 mm out of the position shown on the Design.

214.10 Pile Heads - The tops of all piling shall be cut off to a true plane as shown on the Design or as established by the Designer. Piles shall show a solid head at the plane of cutoff.

Pile heads not encased in concrete shall be protected as follows:

214.10.01 Treated Piles - After cut-off, the pile heads shall be covered with alternate layers of hot roofing asphalt and loosely woven glass fabric, using four applications of asphalt and three layers of fabric.

The fabric shall measure at least 150 mm more in dimension than the diameter of the pile and shall be neatly folded down and secured by large headed galvanized roofing nails or by binding with not less than seven complete turns of galvanized wire securely held in place by large-headed galvanized roofing nails or staples. The edges of the fabric projecting below the fastening shall be neatly trimmed.

214.10.02 Untreated Piles - After cut-off, the pile head shall be thoroughly swabbed with three coats of hot creosote oil and if they are to be capped, they shall also be coated with hot roofing asphalt.

214.11 Not used

214.12 Not used

SECTION 215

BRIDGES

215.01 General - Bridges shall be designed in accordance with the current Canadian Highway Bridge Design Code and the current Ministry standards and practices unless otherwise specified in the Contract. Unless otherwise specified in the Contract, fabrication and erection shall be in accordance with the current Ministry standards and practices and the Design prepared for the particular structure.

SECTION 216

COATING OF STEELWORK

216.1 Scope -This Section covers the operations required for shop, field and/or maintenance coating of structural and miscellaneous steel surfaces. The coating process includes pre-cleaning, surface preparation, coating, and curing of coatings.

216.2 Definitions

216.2.1 Good Painting Practice - The application of generally accepted methods and equipment to clean, contain and coat industrial structures in a safe, practical, environmentally sensitive manner resulting in a long lasting quality protective coating system. These principals are laid out in this Section. Many of the points are drawn from the SSPC (Society for Protective Coatings) publication Steel Structures Painting Manual Volume 1, Good Painting Practice. References are to the current edition unless otherwise stated. These practices are based on experience in the United States and where they conflict with Canadian or British Columbia laws, regulations or guidelines the local regulations shall prevail.

216.2.2 Shop, Field and Maintenance Coating - The series of operations that include the surface preparation, pre-treatment (if required) and application of coatings to structural steel surfaces either in the shop or in field. It also includes the supplying of all labour, equipment and materials. In addition, it encompasses the protection of newly coated surfaces during curing, transport and erection as well as the protection of traffic, property and the environment. Additionally, it also includes the disposal of waste material in accordance with all applicable environmental, fisheries and local regulations and bylaws.

216.2.3 Painting / Coating - The terms painting and coating are considered synonymous for the purposes of this document. Techniques such as metallizing or galvanizing while not typically thought of as painting, are considered here to be painting or coating since they form part or all of a protective system.

216.2.4 Shop Coating - The coating of steel or other components in a shop or plant before shipment to the site of erection. This may include part of a coating system or the whole system including the topcoat.

216.2.5 Field Coating - The coating of steel or components at the place of erection.

216.2.6 Maintenance Coating – The coating of structures in service that have previously been partially or completely coated. The maintenance coating may be one of four general types:

216.2.6.1 Touchup – The cleaning and coating of small selected areas on a structure. Typically, this is used to repair damaged or failed areas of coating. For repairs, sufficient coats are added to bring the new coating up to the level of the existing coating surrounding the repairs. It may also be used as an interim maintenance strategy to extend the lifetime of a coating system overall.

216.2.6.2 Zone Coating – The coating process where specific whole parts of a structure are coated and others are deliberately left. This deals with larger portions of a structure than touchup coating. For example an above deck arch portion of a bridge would be zone coated, but the edges of the flanges would be touched up. Zone coating may use touchup, overcoat, or recoat strategies to meet its objectives. Zone coating also includes coating new steel added to old structures for example steel added for seismic upgrades.

216.2.6.3 Overcoating – The coating of portions of a structure where the structure is touched up, then coated with a topcoat overall. It may include top coating part or all of the structure.

216.2.6.4 Recoating – The coating process where larger portions of old coating on a structure are removed to bare metal and a completely new coating system is applied over the entire area.

216.2.7 Surface Profile - The surface profile is a measure of the roughness of the prepared surface. The profile is normally measured from surface peak to surface valley and expressed as a range (for example 50 to 75 μm or 2 to 3 mils). Besides the depth, the surface profile is dependant on the grit type used to create it. Mineral abrasives such as sand, slag, garnet, etc. produce a 'sharp' profile. Steel shot produces a rounded profile unless mixed with 15% steel grit. Normally the surface profile is specified by both a depth and sharpness parameter.

216.2.8 Containment / Hoarding - The process of encapsulating the work site so that hazardous or nuisance materials cannot escape to the

surrounding environment. SSPC Guide 6 Guide for Containing Debris Generated During Paint Removal Operations specifies types and levels of containment necessary for various cleaning operations.

216.2.9 Ambient Conditions - Ambient conditions refer to such things as wind, relative humidity, dew point, air temperature, substrate temperature etc. that affect the cleaning, application, and curing of coatings. They are very important in obtaining a properly adhered, long lived coating. All coating works shall be within the specified parameters as given in DBSS 216.8.4, DBSS 216.8.5 and DBSS 216.8.9. Where ambient environmental conditions do not fall within these conditions, coating work may continue if the conditions are being met within the containment. This may mean extra hoarding, dehumidification, ventilation and/or temperature control for the duration of cleaning, application and curing of the coating system.

It is worthy of note that some coatings are designed to apply in conditions outside those mentioned above. These coatings and their conditions of use will be individually considered for use.

216.3 Reference Standards - It is recognized that standards are changed or amended from time to time. All specifications mentioned shall be the current editions at the date and time of contract signing unless a specific edition of a reference or standard is stated here or in the Design or Contract.

216.3.1 British Columbia Ministry of Transportation

216.3.1.1 Recognized Products List (RPL) - The Ministry publishes a Recognized Products List of products. This is available on the Ministry website at:

http://www.th.gov.bc.ca/publications/eng_publications/geotech/Recognized_Products_Book.pdf

Only those coating systems listed in DBSS 308 and under the heading of "Coating Systems" in the RPL or as otherwise specified in the Contract shall be used on Ministry Projects. The Ministry may, at its sole discretion, accept alternative materials to fit a unique situation.

216.3.2 SSPC - Society for Protective Coatings - The SSPC publishes two volumes: SSPC Painting Manual Volume 1, Good Painting Practice and SSPC Painting Manual Volume 2, Systems and Specifications, which lay out the basics for painting steel structures. The following SSPC standards and guidelines shall apply unless noted otherwise in the Contract.

216.3.2.1 Surface Preparation Standards

216.3.2.1.1 General - SSPC-SP COM Surface Preparation Commentary for Steel and Concrete Substrates.

216.3.2.1.2 Visual Standards:

Guide to SSPC-VIS 1, Visual Standard for Abrasive Blast Cleaned Steel

Guide to SSPC-VIS 2 Standard Method of Evaluating Degree of Rusting on Painted Steel Surfaces

Guide to SSPC-VIS 3 Visual Standard for Power and Hand Tool Cleaned Steel

Guide to SSPC-VIS 4/NACE No. 7 Guide and Visual Reference Photographs for Steel Cleaned by Water Jetting

SSPC-VIS 5/NACE VIS 9 Guide and Reference Photographs for Steel Surfaces Prepared by Wet Abrasive Blast Cleaning

216.3.2.1.3 Surface Cleaning Standards:

SSPC-SP 1 Solvent Cleaning

SSPC-SP 2 Hand Tool Cleaning

SSPC-SP 3 Power Tool Cleaning

SSPC-SP 5/NACE No. 1 White Metal Blast Cleaning

SSPC-SP 6/NACE No. 3 Commercial Blast Cleaning

SSPC-SP 7/NACE No. 4 Brush-off Blast Cleaning

SSPC-SP 10/NACE No. 2 Near-White Blast Cleaning

SSPC-SP 11 Power Tool Cleaning to Bare Metal

SSCP SP 12/NACE No. 5 Surface Preparation and Cleaning of Steel and Other Hard Materials by High- and Ultrahigh-Pressure Water Jetting Prior to Recoating

SSPC-SP 13/NACE No. 6 Surface Preparation of Concrete

SSPC-SP 14/ NACE No. 8 Industrial Blast Cleaning

SSPC-SP 15 Commercial Grade Power Tool Cleaning

216.3.2.2 Abrasive Specifications:

SSPC-AB 1 Mineral and Slag Abrasives

SSPC-AB 2 Cleanliness of Recycled Ferrous Metallic Abrasives

SSPC-AB 3 Newly Manufactured or Remanufactured Steel Abrasives

216.3.2.3 Paint Application Standards, Guides and Specifications:

SSPC-PA COM Commentary on Paint Application

SSPC-PA 1 Shop, Field and Maintenance Painting of Steel

SSPC-TU 3 Overcoating

SSPC-PA 2 Measurement of Dry Film Coating Thickness with Magnetic Gauges

216.3.2.4 Guides:

SSPC - Guide 6 Guide for Containing Debris Generated During Paint Removal Operations

SSPC – Guide 12 Guide for Illumination of Industrial Painting Projects

216.3.2.5 Technical Updates:

SSPC-TU 4 Field Methods for Retrieval and Analysis of Soluble Salts on Substrates

SSPC-TU 6 Chemical Stripping of Organic Coatings from Steel Surfaces

SSPC-TU 7 Conducting Ambient Air, Soil, and Water Sampling During Surface Preparation and Paint Disturbance Activities

216.3.2.6 Qualification Procedures - These requirements will be laid out specifically in the Contract for each project and will depend primarily on the work to be done. Contractors may require current certification in one or more of the following levels to perform Work as called for in the Contract.

SSPC-QP COM Commentary on Qualification Procedures

SSPC-QP 1 Standard Procedure for Evaluating Painting Contractors

SSPC-QP 2 Standard Procedure for Evaluation Painting Contractors to Remove Hazardous Paint

SSPC-QP 3 Standard Procedure for Evaluating the Qualifications of Shop Painting Contractors

NACE Coating Inspection Program Level 2 with Peer Review

2.16.3.3 Hot Dip Galvanizing

CAN/CSA-G164, Hot Dip Galvanizing of Irregularly Shaped Articles

216.3.4 Metallizing

CAN/CSA-G189 Sprayed Metal Coatings for Atmospheric Corrosion Protection

216.3.5 Colour Standards - Refer to US Federal Specification 595B Colors. Topcoat colours shall be in accordance with the Design or as determined by the Ministry Representative.

216.4 Environmental Considerations

216.4.1 General - All coating works including preparation shall comply with DBSS 165, Protection of the Environment.

216.4.2 Environmental Agencies Affected - Depending on the Work, some or all of the following Environmental Agencies and other affected parties have interests and may need to be contacted for their input to the Work. The list below is not exhaustive.

216.4.2.1 Federal:

Fisheries and Oceans Canada

Environment Canada

Transport Canada

Port Authorities

216.4.2.2 British Columbia:

Ministry of Water Land and Air Protection (WLAP)

Provincial, Municipal or Private Utilities if Affected

Railways if Affected

Aboriginal Bands if Affected

Water User Groups if Affected

Other Affected Parties, as necessary

The various parties' interests vary from legislated involvement to public or private interest in the

portion of the environment potentially affected by the coating project. It is the Contractor's responsibility to liaise with all the affected parties and to meet the requirements of the relevant authorities.

216.5 Containment of Coating Operations

216.5.1 General - Full containment shall be provided to prevent contamination of the environment. This enables work to be accomplished by physically limiting the spread of debris and facilitates cleanup. Current containment technology is able to contain virtually all debris generated by washing, water jetting, hand, power tool, or abrasive blasting, and coating operations.

216.5.2 SSPC - Guide 6 - Containment shall be provided in accordance with SSPC - Guide 6, Guide for Containing Debris Generated During Paint Removal Operations. In general, the Ministry wishes to minimize the environmental impact of all Ministry coating work. To this end, all coating works shall be fully enclosed to prevent contamination of the environment. The coating specification will contain specific language to exactly determine the desired containment end performance requirements. Containment shall be approved by the pertinent WCB, Environment and Fisheries Authorities.

216.5.3 Containment of Debris and Disposal of Debris - All debris from cleaning and coating operations must be contained in such a way as not to contaminate the environment. All debris shall be contained, tested and disposed of in accordance with current environmental, fisheries and local regulations. The Contractor shall provide written proof of disposal of all materials including way-bills and transport logs.

216.6 Construction Site Safety

216.6.1 General - General site safety is governed primarily by DBSS 135 Construction Site Safety. Contractor shall also meet all WCB regulation requirements for work undertaken.

216.6.2 Specific Safety Issues Relating to Bridges and Coating Work

216.6.2.1 Working at Height - All work occurring at height shall conform to all WCB regulation requirements.

216.6.2.2 Lead Paint - Many of the Ministry structures are painted with paint containing lead. While the Ministry will attempt to indicate the presence of paint containing lead, it is the responsibility of the Contractor to ascertain the presence and extent of lead containing paint if any; and to conform to all regulations for work undertaken. This includes but is not necessarily limited to WCB and WLAP regulation requirements for working with, storing, transporting, disposing of and documenting any work process involving lead paint.

216.6.2.3 Enclosed Spaces - Many of the Ministry structures contain enclosed spaces or spaces that the WCB defines as enclosed spaces. In some instances the hoarding or containment is considered an enclosed space. It is the responsibility of the Contractor to ascertain whether or not a space is defined as an enclosed space and to adjust their work procedures to meet all regulation requirements.

216.7 Surface Preparation

216.7.1 General - The useful lifetime of a coating is extended by appropriate surface preparation prior to coating. The surface preparation shall be in accordance with the Contract requirements. The Designer will define the quality of surface preparation for a given area to be coated. Where this surface preparation differs from the coating manufacturer's recommended surface preparation, the Design shall be considered correct. However, in the case where the coating manufacturer's specification requires more stringent surface preparation, then the coating manufacturer's specification shall apply unless otherwise approved by the Ministry Representative.

Where surface preparation is not defined in the Contract or in the Design, the manufacturer's specifications shall be used.

216.7.2 Washing

216.7.2.1 General - Washing is a procedure for removing surface contaminants including oil, grease, drawing and cutting compounds, soil, salts, etc. from either steel or coated surfaces. Removal of the contaminants shall be accomplished by the use of any one or a combination of the following: potable water, alkaline cleaners, solvents, emulsion cleaners, steam cleaning (with detergents), high pressure

water, scrubbing (with detergent) or other approved method.

Alkaline cleaners, detergents, solvents, etc. shall not be used over streams, lakes, rivers, etc. or agricultural land without first obtaining approval from the Ministry of Water, Land and Air Protection (WLAP) as well as Fisheries and Oceans Canada and Transport Canada.

The washing procedure shall be conducted so that any part of the structure which has already been cleaned, or newly coated, shall not be contaminated with the cleaning chemicals or solvents.

Immediately after treatment with cleaning chemicals, the surfaces shall be thoroughly washed with clean water to remove detrimental residues.

216.7.2.2 Residual Ion Levels - One main reason for washing is to remove residual anions such as chloride and sulphate as well as the ferrous cation and other non-visible contaminants. Currently, low risk levels for most of the ions are not well defined, however chloride levels are. Generally levels shall be at or below the SSPC-SP12 NV-2 level, less than 7 µg/cm2 chloride ion, 10 µg/cm2 soluble ferrous ion, and 17 µg/cm2 sulfate, immediately prior to application of coatings when tested by SSPC-TU 4 swabbing extraction method.

Chloride levels shall be measured using one of the methods in SSPC-TU 4, and calculated using the formula in Appendix C of that specification. Verification of NV-2 level shall be done prior to surface preparation operations and prior to primer application.

216.7.3 Water Cleaning/Jetting - Water jetting encompasses low pressure water cleaning (LP WC), high pressure water cleaning (HP WC), high pressure water jetting (HP WJ), and ultrahigh pressure water jetting (UHP WJ).

SSPC-SP12/NACE No. 5, Surface Preparation and Cleaning of Steel and Other Hard Materials by High- and Ultrahigh- Pressure Water Jetting Prior to Recoating defines specifications for visual and non-visual surface preparation by water jetting.

Water jetting is inherently dangerous and safety precautions as listed in SSPC-SP 12 and in SSPC Steel Structures Painting Manual, Volume 1, Good Painting Practice, shall be followed. Where current WCB regulations conflict with SSPC-SP12/NACE No. 5, the WCB regulations shall prevail. Water used in water jetting units shall be clean and free of contaminants that may leave deposits on the surface being cleaned or affect the adhesion of subsequent coatings applied. Water supply to the water jetting units must meet regulations for water taken from streams if on-site water from streams is used.

All waste water and coating debris shall be contained and disposed of in accordance with Environmental and Fisheries Agencies or other applicable regulations.

Wastewater must meet regulations for disposal on-site if it is to be disposed of on-site. These regulations are typically those of Fisheries and Oceans and WLAP, but may not be limited to those governmental agencies. As well, DBSS 165 Protection of the Environment, lays out the Contractor Responsibilities.

216.7.4 Manual and Power Tool Cleaning - Manual cleaning is a procedure for removing loose mill scale, loose rust, loose coating and other detrimental foreign matter by chipping, scraping, grinding, and wire brushing, etc. The tools used for this method of cleaning may be either hand or power tools; they shall be of high quality, and maintained in good repair.

The standard of cleaning obtained by this method shall be SSPC-SP 2, SP 3, SP 11or SP 15 as specified in the Design. Where surface preparation is not defined in the Design, the manufacturer's specifications shall be used.

In specific circumstances the methods available may be limited by the Design.

In all four preparation methods listed above, the surface shall be cleaned of dirt, grease and soluble salts prior to treatment. Heavier layers of rust shall be removed by chipping, before any of the other cleaning tools are used.

Finally, all traces of dust shall be removed prior to coating.

216.7.5 Abrasive Blast Cleaning - Abrasive blast cleaning is a procedure for removing mill scale, rust, rust scale, paint or other foreign matter by use of abrasives propelled through nozzles or by centrifugal wheels. It is also used to roughen the surface of the substrate to enhance adhesion.

If compressed air blasting is employed, proper line filters and dryers, which remove all oil and water from the air, shall be used. Compressed air shall be periodically tested for presence of these contaminants.

The standard of cleaning obtained by this method shall be SSPC SP 5, SP 6, SP 7, SP 10, or SP 14

In all preparation standards, it is assumed that prior to treatment the surface has been cleaned of dirt, oil, grease and soluble salts etc. If necessary, heavier layers of rust shall be removed by chipping before the blast cleaning operation commences.

A cleaned and blasted surface shall be coated within a maximum of 8 hours of cleaning and before the surface shows evidence of rust blooming. If either of these criteria is not met, then the surface shall be re-prepared to specification. If a prepared surface is contaminated before coating, it shall be re-cleaned and/or re-blasted in accordance with the Design or manufacture's specifications before coating is applied.

Finally, all traces of dust and sand, etc. shall be removed prior to coating.

216.7.6 Abrasive Materials - Abrasive materials, unless specified, must be accepted by the Quality Manager and Ministry Representative.

216.7.7 Recycled Steel Abrasive - Recycled steel abrasives are permitted, however the working material must meet SSPC-AB 2 requirements and must contain a minimum 15% steel grit to produce an angular/sharp profile.

216.7.8 Containment of Debris and Disposal of Debris - All debris from cleaning and coating operations must be contained in such a way as not to contaminate the environment. All debris shall be contained, tested and disposed of in accordance with current environmental, fisheries and local regulations. The Contractor shall provide written proof of disposal of all materials including way-bills and transport logs.

216.8 Application of Coating Materials

216.8.1 Pretreatment - Newly cleaned surfaces may require pretreatment other than washing or abrasive blasting prior to the application of the prime coat of paint. The application of the pretreatment shall be strictly in accordance with the manufacturer's directions. When pretreatments are used, particular care shall be taken to prevent contamination of adjacent surfaces.

216.8.2 General - In general, Contractors shall follow SSPC-PA 1 Shop, Field, and Maintenance Painting of Steel when applying coatings. Where DBSS 216 and SSPC-PA 1 disagree, DBSS 216 shall be taken as applicable.

216.8.3 Materials Storage and Handling - Materials handling and use shall be as per SSPC-PA 1, 5.1. Coatings shall be stored in a secure building on-site, but not on the structure itself. Coatings taken onto the structure in cans, pails or totes shall be contained so that should they spill, no coating will escape to contaminate the adjacent area or environment.

216.8.4 Temperature - Coatings shall only be applied when the ambient, surface or coating temperature is within the range indicated by the Manufacturer's written instructions. Coating shall not be applied when the substrate temperature is less than 3°C above the Dew point. Winter or quick dry catalysts or coating formulations shall not be used. Some coatings are designed for use in marginal conditions. Their use shall be as specified by the Designer and shall be acceptable to the Ministry Representative. In unique situations where coatings must be used outside these limits, they shall only be used as specified by the Designer and when permitted by the Ministry Representative. Lack of permission may necessitate removal, re-cleaning and reapplication by the Contractor.

216.8.5 Humidity - Coatings shall not be applied in wet weather, fog, mist, rain, snow and/or relative humidity in excess of 85%. Coatings shall not be applied to wet or damp or icy surfaces and care shall be taken to ensure that any overnight dew has completely evaporated before coating commences in the morning. Some coatings are designed to cure in higher humidity environments and may be specified on the Design.

216.8.6 Mixing and Thinning - All ingredients in any container of paint shall be thoroughly mixed before use and shall be agitated often enough during application to keep the paint in uniform condition. In all cases of heavily pigmented paints (e.g. "Zinc Rich" paints) the material shall be continuously agitated during application.

Mixing shall be done in accordance with the manufacturer's instructions and in accordance with good painting practice as outlined in SSPC Steel Structures Painting Manual, Volume 1, Good Painting Practice and SSPC-PA 1.

Thinner shall only be added with the written consent of the Manufacturer and only after it is shown that satisfactory application cannot be obtained with proper adjustment of the spray equipment and air pressures. In no case shall more than 12.5% of thinner by volume be added to a paint.

Only thinner specified and supplied by the Manufacturer for that product shall be added to a product. Thinners used for cleanup shall not be recycled as thinning agents. Paint shall be mixed by mechanical methods. Paint shall be mixed in a manner that will ensure breaking up of all lumps, complete dispersion of settled pigment and a uniform composition. After mixing, all paint shall be strained to remove skins and other undesirable matter; the strainer shall not be so fine as to remove any pigment.

Thinner, when approved, shall be added after initial mixing is completed. The quantity of thinner added shall be measured in and thoroughly mixed until the paint is again of uniform consistency.

216.8.7 Induction Time - Manufacturers' induction times shall be strictly observed before the coating is applied. Since induction times are a function of temperature, temperatures outside the normal range may necessitate changing the induction period. Revised times if necessary shall be provided by the Coating Manufacturer in writing and subject to acceptance by the Quality Manager and Ministry Representative.

216.8.8 Pot Life - Manufacturer's estimates of pot life shall be adhered to and no coating shall be used after the recommended pot life has expired.

Since pot life is a function of temperature, in hot or cold conditions the pot life may have to be modified reflect that temperature. Revised pot life times if necessary shall be provided by the Coating Manufacturer in writing and subject to acceptance by the Quality Manager and Ministry Representative.

216.8.9 Wind - Paint shall not be sprayed when the wind speed is over 15 km per hour. This includes gusts. Wind speed shall be measured where the work is taking place, for example within the containment.

216.8.10 Curing Conditions - Coating lifetime may be affected by curing conditions. Coatings shall be protected from rain during their curing. Coatings shall be cured under the Coating Manufacturer's stated conditions. This may require heating, cooling, humidity or ventilation adjustments to bring the ambient conditions into line with the Coating Manufacturer's recommended conditions for curing. In enclosed spaces, ventilation must be provided to remove evaporating vehicle from the coating.

Failure to achieve proper curing or curing conditions may necessitate removal, re-cleaning, reapplication or repair of the coating by the Contractor.

216.9 Inspection

216.9.1 General - The Contractor shall provide the quality control and quality assurance functions for the Work. Quality audit will be done by the Ministry. The Contractor shall produce a quality plan for all work to be done and submit this plan for the Ministry's acceptance. The Ministry will audit the quality process via the Ministry Representative.

216.9.2 Quality Control and Quality Assurance Inspection - Quality Control and Quality Assurance inspectors shall be qualified NACE Level 2 plus peer review, i.e. NACE Certified Coating Inspectors. Inspection records produced shall be quickly forwarded to the Quality Manager and Ministry Representative. The records shall be electronic files, preferably Microsoft Excel files, and shall be forwarded to the Quality Manager and Ministry Representative in that format to speed dissemination and aid any data manipulation required for the quality audit. Where this is not possible due to lack of communication technology, the records may be hard copy or hand delivered electronic files. The Quality Control and Quality Assurance inspector(s) shall follow the quality plan as developed by the Quality Manager and accepted by the Ministry.

All washing and cleaning work shall be inspected by the Contractor before any coating is applied. Each coat of paint shall be inspected by the Contractor and passed before the succeeding coat is applied.

Coated surfaces rejected shall be made to comply with all requirements of this specification.

216.9.2.1 Hold points for inspection shall include, but may not be limited to:

Installation of scaffolding and other accesses

Installation of containment

Surface after washing

Surface after cleaning prior to coating application

Surface after Prime coat

Surface after Penetrating Sealer if any

Surface after Stripe Coat

Surface after each midcoat

Surface after caulking

Surface after topcoat

Removal of containment

Removal of scaffolding and other accesses

216.9.2.2 Accessibility for Inspection - The Contractor shall provide the Quality Manager and Ministry Representative safe access to the specific areas of the jobsite under inspection. In order to limit the work disruption, the Contractor shall notify the Quality Manager and Ministry Representative of upcoming hold points at least 24 hours prior to the hold point. The Ministry Representative may perform quality audit inspections in conjunction with or subsequent to the Contractor's quality control and quality assurance inspections. The Ministry Representative will attempt to limit delays to the Contractor as much as possible, but retains the right to obtain as much information from the quality audit inspections as needed to justify acceptance or rejection of the work.

The Contractor shall maintain a minimum 200 foot-candles illumination at all areas of the work to be inspected. This may include adding lighting to augment natural light. Lighting if added shall be according to SSPC Guide 12, Guide for Illumination of Industrial Painting Projects.

216.9.3 Thickness Measurement - Large areas amenable to the application of SSPC-PA 2 Measurement of Dry Paint Thickness with Magnetic Gauges shall be measured by this method. The use of PA 2 is not considered to limit the number of measurements should additional readings be needed for any reason.

The Contractor shall set aside a suitable area of steel on the structure to be coated and dry abrasive blast clean the area with the Contractor's equipment and blast abrasive to the standard required by the coating specification. This area shall be protected from moisture and shall be used as a standard for acceptable surface cleanliness and anchor profile. A portion of the area shall also be used as an area to verify the calibration of dry film thickness (DFT) gauges. It is recognized that the magnetic plane on the blast cleaned surface is above the bottom of the profile. The Contractor shall fill that difference with coating. If requested by the Quality Manager or Ministry Representative, the Contractor shall produce and calibrate a secondary, portable calibration standard for dry film thickness gauges. The dry film thickness of small and/or more complex areas shall be measured as agreed upon by the Contractor's Inspector and the Ministry Representative. Gauges are to be calibrated according to SSPC-PA 2 at the beginning and end of each work shift.

216.9.4 Tinting Multilayer Systems - Each separate coat, penetrant, stripe, midcoat(s) and topcoat shall be coloured or tinted to differentiate it visually from the coats above and below. This is an aid to application and inspection and also serves as an indicator of wear for the life of the coating.

216.9.5 Coating Systems - DBSS 308 lists coating systems used or identified by the Ministry and suggests situations where the systems are best applied. Coating systems meeting DBSS 308 requirements are listed in the Recognized Products List. Only specified coating systems from the Recognized Products List shall be used.

Occasionally, special requirements mandate the use of coating materials not already listed in DBSS 308. These requirements will be specified in the Contract or will be shown on the Design. Approval from the Ministry Representative is required for the use of coating systems that are not listed on the Recognized Products List.

216.9.6 Coatings from Different Manufacturers - The Contractor shall not mix or interchange coatings or solvents from different manufacturers.

216.9.7 Coating Flaws - A coat of paint will only be accepted if none of the following faults are apparent:

- Coatings showing sags, runs, holidays, pinholes, orange peel, dry spray, shadowing or other application defects.
- Poor coverage at rivet heads, bolts, threads, plate edges, crevices, pockets, corners or re-entrant angles.
- Surfaces damaged mechanically by moving scaffolding, equipment, etc. or damage by rain, flying dust, etc. or damage from any other cause.

Sags and runs may be repaired or treated as per SSPC-PA 1, Section 7.4.5

Coatings shall be applied as per SSPC-PA 1, except where directed otherwise by this specification.

216.9.8 Rejection - Areas not meeting specification shall be repaired by the Contractor. At the discretion of the Quality Manager and the Ministry Representative, small areas may be spot repaired. Larger areas shall be re-cleaned to specification and the coating built up again to specification.

216.10 Application Methods (Excluding metallizing and Hot Dip Galvanizing)

216.10.1 General - Coatings are designed to be applied by specific application techniques. These techniques are specified by the Manufacturer to ensure that the risk of failure is minimized. Manufacturer's product application sheets shall be followed when applying coatings.

Where an application method is not possible due to circumstances at a particular structure, the Contractor shall obtain written instructions from the Manufacturer for a new application method. The Quality Manager and Ministry Representative must accept such methods before they are used.

216.10.2 Brush or Roller Application - Brushes or rollers shall be of high quality and of a style and material that will enable proper coating application. The paint shall be brushed or rolled so that there are no runs or sags, minimum brush marks and a uniform thickness. Achieving the specified thickness for a coating may require more than one coat with brush or roller. The Contractor is responsible for achieving the specified thickness.

216.10.3 Air Spray Application - The equipment shall be capable of properly atomizing the paint to be applied. The equipment shall be kept clean and maintained in excellent condition. Equipment that has dirty or malfunctioning parts shall not be used until the parts are cleaned, repaired or replaced as necessary, to the satisfaction of the Quality Manager and Ministry Representative.

Paint shall be applied in a uniform layer, with overlapping at the edge of the spray pattern. The spray pattern shall be adjusted so that the paint is deposited uniformly. During application, the gun shall be held perpendicular to the surface and at a distance that will ensure that a wet layer of paint is deposited on the surface. The trigger of the gun shall be released at the end of each stroke.

All runs and sags shall be brushed out immediately, or the paint shall be removed and the surface repainted.

Areas inaccessible by spray such as cracks and crevices and the blind sides of rivets and bolts shall be painted by brush. Stripe coat material may be applied by spray to speed up the transfer of material to the surface, but all stripe coating must be brushed in to ensure complete wetting and coverage.

The air caps, nozzles and needles shall be those recommended by the manufacturers of both the material being sprayed and the equipment being used.

Proper air line filters and dryers shall be provided to remove oil and condensed water from the air. The air from the spray gun impinging against the surface shall show no condensed water or oils.

The pressure on the material in the pot and of the atomizing air shall be adjusted for optimum spraying effectiveness. Both material and atomization pressures shall be controlled by suitable pressure regulators with gauges. The atomizing air pressure shall be high enough to properly atomize the paint, but not so high as to cause excessive fogging of paint, excessive evaporation of solvent or loss by overspray.

The air compressor used shall be of sufficient size to maintain the regulated air pressures constant at all times, irrespective of the amount of equipment that is being operated from the compressor.

216.10.4 Airless Spray Application - The equipment shall be capable of properly atomizing the paint to be applied. The equipment shall be